Addition of 2 number

$(4789)_{16} = (18313)_{10}$

$(6488)_{16} = (25736)_{10}$

$= (44049)_{10}$

$= (AC11)_{16}$

Assembly Code	Machine	Comment
MOV AX, 0	B8 00 00	
SAHF	9E	
ADD AX, 4789H	05 **8947**	
ADC AX, 6488H	15 **8864**	ADC stands for Add with Carry. It adds an immediate value plus the carry flag (CF) to AX (Adds 6488H + CF to AX).

- SAHF stands for Store AH into Flags.
- It copies the contents of the AH register into the lower 8 bits of the FLAGS register (specifically, SF, ZF, AF, PF, and CF).

Step:

1. AD
2. DA
3. Machine value insert
4. Got to start position (- Press)
5. STP (For Run)
6. + (Verify)
7. Continuously do step 5 &6 until the result is displayed.

Subtraction of 2 number

$(4789)_{16} = (18313)_{10}$

$(3567)_{16} = (13671)_{10}$

$= (4642)_{10}$

$= (1222)_{16}$

Assembly Code	Machine	Comment
MOV AX, 0	B8 00 00	
SAHF	9E	
ADD AX, 4789H	05 **8947**	
SUB AX, 3567H	2D **6735**	

Multiplication of 2 number

Assembly Code	Machine	Comment
MOV AX, 0	B8 00 00	
SAHF	9E	
MOV AX, 02	B0 02	
MOV BX, 02	B3 02	
MUL BX	F6 E3	

Division of 2 number

Assembly Code	Machine	Comment
MOV AX, 0	B8 00 00	
SAHF	9E	
MOV AX, 0008H	B8 0800	
MOV BL, 02H	B3 02	
DIV BL	F6 F3	

Increment

Assembly Code	Machine	Comment
MOV AX, 0	B8 00 00	
SAHF	9E	
ADD AX, 0001H	05 0100	
INC AX	FE C0	

Decrement

Assembly Code	Machine	Comment
MOV AX, 0	B8 00 00	
SAHF	9E	
ADD AX, 0002H	05 0200	
DEC AX	FE C8	

NEG (2's Complement)

0000 0010 = 02

1111 1110 = FE

Assembly Code	Machine	Comment
MOV AX, 0	B8 00 00	
SAHF	9E	
ADD AX, 0002H	05 0200	
NEG AX	F6 D8	